Siddhi Sanjay More

Mechanical Systems · Embedded Systems · Robotics

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EDUCATION

Northeastern University **Expected May 2027**
MS in Robotics, Concentration: Mechanical Engineering | GPA: 3.25 | Global Student Award
Courses: Robot Sensing and Navigation, Robotics Mechanics and Control, Wearable Robotics, Control Systems Engineering

Mumbai University, India **Jul 2020 - Jul 2024**
BE in Mechanical Engineering, Honors: Robotics
Courses: FEA, Machine Design, Automation and AI, CAD/CAM, IoT, Supply Chain Management

TECHNICAL SKILLS

Mechanical and Fabrication: Mechanical assemblies, structural analysis, materials selection, tolerancing, prototyping, failure analysis, laser and foam cutting, FDM (Ultimaker), SLA resin, soldering (SMD and through-hole), PCB milling, hand and power tools
CAD and Analysis: SolidWorks, Fusion 360, CATIA, AutoCAD, NX 11, ANSYS Workbench, Creo, KiCad, EasyEDA, DOE and ANOVA
Embedded Systems: ESP32-S2, Arduino, MPU6050, FSR, CAN-FD, I2C, L298N, microcontrollers, real-time systems, firmware development, hardware debugging
Robotics and Software: ROS/ROS2, Nav2, SLAM, impedance control, gait phase estimation, Moteus C1, GCC-PHAT TDOA, Python, C, MATLAB/Simulink, Stateflow, Embedded Coder, TensorFlow/Keras, Linux, Git

PROJECTS

Smart Ankle Prosthetics, *Northeastern University* **Jan 2026 - Present**
- Built a Simulink/Stateflow gait phase estimator classifying heel-strike, mid-stance, toe-off, and swing from fused IMU/FSR data; predicts next phase 75 ms ahead to pre-position the ankle
- Instrumented the intact limb with MPU6050 and FSR sensors on an ESP32-S2 at 200 Hz; logged synchronized streams to CSV via Python
- Architected CAN-FD communication to a Moteus C1 motor controller transmitting phase suggestions rather than direct commands

Sound Triangulation & Doppler Detection, *Northeastern University* **Sep 2025 - Dec 2025**
- Implemented GCC-PHAT TDOA localization with a microphone array to classify hazard sounds and resolve vehicle approach direction within ±10° at under 100 ms latency

Semi-Autonomous Aquatic Cleanup Robot, *Smart India Hackathon* **Jul 2023 - Oct 2023**
- Designed chassis, hull, and crusher mechanism in SolidWorks; ran von Mises FEA to validate structural integrity before fabrication
- Trained a CNN (TensorFlow/Keras) on 991 images for water hyacinth classification, reaching 87.5% test accuracy

Trajectory Generation for SLAM with ROS2 and TurtleBot3, *Northeastern University* **Jan 2025 - Apr 2025**
- Built a ROS2 control package with Cartographer SLAM and Nav2 that replaced manual teleoperation with automated closed-loop trajectory execution, achieving 0.053 m translation error across 10 spawn poses

EXPERIENCE

IIT Bombay and 1Stop.ai, *Robotics and CAD Intern, Mumbai* **Feb 2022 - Apr 2022**
- Derived DH parameters and built forward and inverse kinematic solvers for a 4-DOF SCARA robot in MATLAB; validated motion profiles in Robo-Analyzer and achieved sub-mm end-effector accuracy across the full workspace
- Ran static structural analysis in ANSYS Workbench under max-payload load cases; identified a stress concentration that triggered a design revision before fabrication
- Designed mechanical parts in SolidWorks against client specifications, applying GD&T and manufacturing tolerancing standards throughout

SAE Aero Design West, *Design member, Mumbai, India* **Aug 2021 - Nov 2021**
- Designed landing gear and wings in SolidWorks, ran FEA to validate structural integrity, and sourced materials and operated laser cutters for fabrication

PUBLICATIONS AND ENGAGEMENT

- "Experimentation in Injection Molding: Simulation and Parameter Optimization Using DOE and ANOVA" AIP Conference Proceedings, ICAPSM 2024
- Design Head, Rotaract Club of CRCE | Managed graphic design and social media for club events; coordinated sponsorship outreach and communications with external partners (Aug 2021 to Jul 2024)